

Taylor Expansions

Single & multivariable · the formulas, the toolkit, the pictures, and fully worked examples

1 The core idea

Near a point a , a smooth function looks like a polynomial. Match value, slope, curvature, ... one derivative at a time:

Taylor's theorem (one variable)

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}_{T_n(x)} + R_n(x)$$

Lagrange remainder: for some ζ between a and x ,

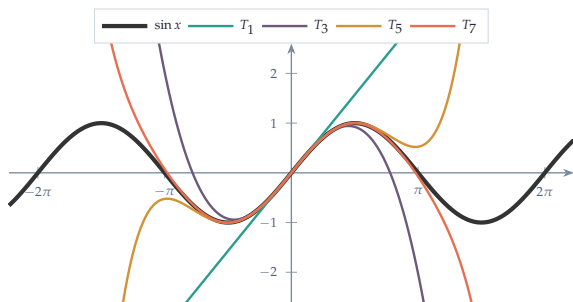
$$R_n(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} (x-a)^{n+1}$$

Error bound: if $|f^{(n+1)}| \leq M$ on the interval,

$$|R_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}.$$

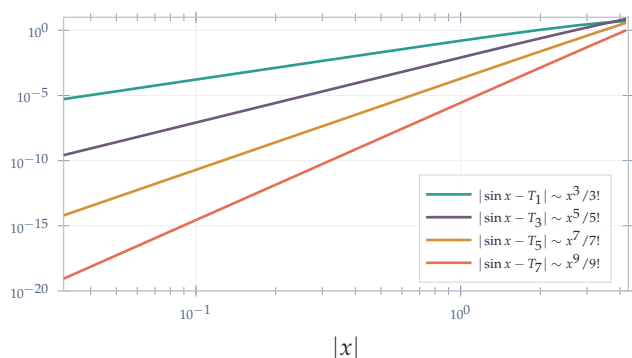
Peano form: $R_n(x) = o((x-a)^n)$ as $x \rightarrow a$. **Maclaurin:** the case $a = 0$.

Why the $k!$? Differentiating $(x-a)^k$ k times gives $k!$; dividing by it makes $T_n^{(k)}(a) = f^{(k)}(a)$ for every $k \leq n$.



Each new term hugs $\sin x$ over a wider window. Accuracy is *local*: great near a , then it peels away.

2 How fast does the error shrink?



Log-log axes: the error of T_n is a straight line whose slope is the first omitted power. Halve x and the T_7 error drops $2^9 \approx 500 \times$. Far from a , all bets are off.

Symmetry saves work. f odd $\Rightarrow$ only odd powers ($\sin, \tan, \arctan, \sinh$); f even $\Rightarrow$ only even powers ($\cos, \cosh, e^{-x^2}$). So for even f , $T_{2m} = T_{2m+1}$: a free extra order of accuracy in the error bound.

3 The standard series (memorise these)

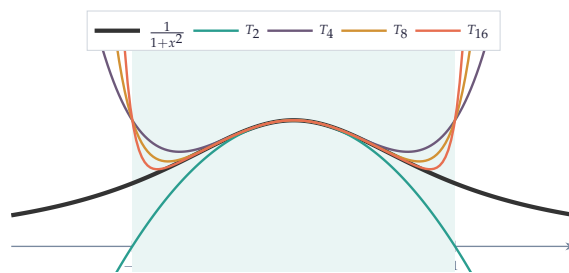
$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots & \text{all } x \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots & \text{all } x \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots & \text{all } x \\ \frac{1}{1-x} &= 1 + x + x^2 + x^3 + \dots & |x| < 1 \\ \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots & -1 < x \leq 1 \\ (1+x)^\alpha &= 1 + \alpha x + \frac{(\alpha)}{2} x^2 + \frac{(\alpha)}{3} x^3 + \dots & |x| < 1 \\ \arctan x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \dots & |x| \leq 1 \\ \tan x &= x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots & |x| < \frac{\pi}{2} \\ \sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots & \text{all } x \\ \cosh x &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots & \text{all } x \\ \arcsin x &= x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots & |x| \leq 1 \end{aligned}$$

Useful binomials: $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots$, $\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \dots$, where $\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}$.

4 Radius of convergence

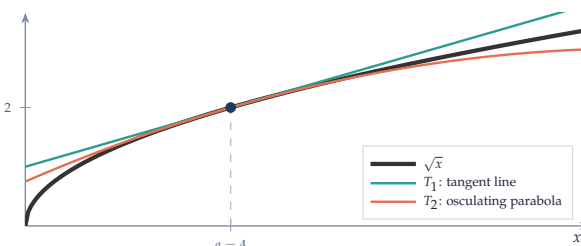
Where does the series actually work?

The series about a converges on $|x-a| < R$, where R is the distance from a to the nearest **complex** singularity of f . Ratio test: $R = \lim |c_k/c_{k+1}|$ when it exists.



$1/(1+x^2)$ is perfectly smooth on $\mathbb{R}$, yet the series dies at $|x|=1$: the poles at $x = \pm i$ are distance 1 from 0.

5 The geometry of orders



T_0 matches height, T_1 slope, T_2 bending. See Example 1.4.

6 The toolkit: never differentiate if you can avoid it

Direct differentiation works but is slow and error-prone past order 2. Build new series from the standard ones:

- 1. Substitute:** $e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \dots$ (replace x by $-x^2$).
- 2. Multiply:** truncate each factor to the order you need, expand, discard higher terms.
- 3. Divide:** write $f/g = \sum c_k x^k$, multiply through by g , match coefficients.
- 4. Compose:** $f(g(x))$ with $g(0) = 0$: plug the series of g into f .
- 5. Differentiate / integrate** term by term inside the radius: $\arctan x = \int_0^x \frac{dt}{1+t^2}$.
- 6. Shift the centre:** for $a \neq 0$, put $x = a + h$ and expand in h .

Order bookkeeping. To get the product correct to x^n : if g starts at x^p , you only need f to order x^{n-p} and vice versa. For $f(g(x))$ with $g = O(x^p)$, you need powers g^k only up to $k = \lfloor n/p \rfloor$.

Example 1.1 Multiplication: $e^x \cos x$ to $O(x^5)$

Truncate $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$, $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24}$.

Collect by power

$$\begin{aligned} x^0: & 1 \\ x^1: & 1 \\ x^2: & \frac{1}{2} - \frac{1}{2} = 0 \\ x^3: & \frac{1}{6} - \frac{1}{2} = -\frac{1}{3} \\ x^4: & \frac{1}{24} - \frac{1}{4} + \frac{1}{24} = -\frac{1}{6} \end{aligned}$$

Result $e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)$

Example 1.2 Division: $\tan x$ to $O(x^7)$

$\tan x$ is odd, so set $\tan x = c_1 x + c_3 x^3 + c_5 x^5$ and use $\tan x \cdot \cos x = \sin x$:

$$(c_1 x + c_3 x^3 + c_5 x^5) \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) = x - \frac{x^3}{6} + \frac{x^5}{120}$$

Match x : $c_1 = 1$. x^3 : $c_3 - \frac{1}{2} = -\frac{1}{6} \Rightarrow c_3 = \frac{1}{3}$.
 x^5 : $c_5 - \frac{c_3}{2} + \frac{c_1}{24} = \frac{1}{120} \Rightarrow c_5 = \frac{1}{120} + \frac{1}{6} - \frac{1}{24} = \frac{2}{15}$.

Result $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + O(x^7)$

Example 1.3 Composition: $e^{\sin x}$ to $O(x^5)$

Let $u = \sin x = x - \frac{x^3}{6} + O(x^5)$. Since $u = O(x)$, keep e^u to u^4 :

$$u^2 = x^2 - \frac{x^4}{3} + \dots, \quad u^3 = x^3 + \dots, \quad u^4 = x^4 + \dots$$

$$\begin{aligned} e^u &= 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \frac{u^4}{24} \\ &= 1 + \left(x - \frac{x^3}{6}\right) + \left(\frac{x^2}{2} - \frac{x^4}{6}\right) + \frac{x^3}{6} + \frac{x^4}{24} \end{aligned}$$

Result $e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + O(x^5)$

Note the x^3 terms cancel exactly.

Example 1.4 Non-zero centre + error bound: $\sqrt{4.1}$

Derivatives at $a = 4$ $f = \sqrt{x}$: $f(4) = 2$, $f' = \frac{1}{2\sqrt{x}} \rightarrow \frac{1}{4}$,
 $f'' = -\frac{1}{4x^{3/2}} \rightarrow -\frac{1}{32}$.

$$T_2(x) = 2 + \frac{x-4}{4} - \frac{(x-4)^2}{64}$$

Evaluate $T_2(4.1) = 2 + 0.025 - 0.00015625 = 2.02484375$

Bound $f''' = \frac{3}{8x^{5/2}}$ is decreasing, so $M = f'''(4) = \frac{3}{256}$:

$$|R_2| \leq \frac{3}{256} \cdot \frac{(0.1)^3}{3!} \approx 1.95 \times 10^{-6}$$

Check $\sqrt{4.1} = 2.0248456\dots$; true error 1.92×10^{-6} . ✓

Example 1.5 Limits: $\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^3}$

$e^x \sin x = (1 + x + \frac{x^2}{2})(x - \frac{x^3}{6}) + O(x^4) = x + x^2 + (\frac{1}{2} - \frac{1}{6})x^3 + O(x^4)$.

Numerator $= \frac{x^3}{3} + O(x^4)$, so the limit is $\frac{1}{3}$.

Rule: expand the numerator until the first term that survives cancellation; that order must match the denominator.

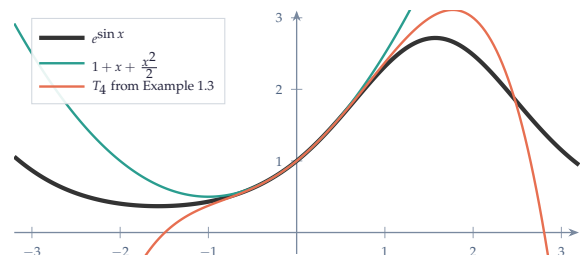
Example 1.6 Integration: $\int_0^1 e^{-x^2} dx$

Substitute then integrate term by term:

$$\int_0^1 \sum_{n \geq 0} \frac{(-1)^n x^{2n}}{n!} dx = \sum_{n \geq 0} \frac{(-1)^n}{n! (2n+1)}$$

$= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots \approx 0.74749$ (five terms).

Alternating series: error < next term $= \frac{1}{1320} \approx 7.6 \times 10^{-4}$.
 True value 0.746824... ✓



The $-x^4/8$ term (coral) bends the curve back down and stays close to $e^{\sin x}$ out to $|x| \approx 1$; the quadratic (teal) overshoots much sooner.

Classic mistakes

- ▶ Truncating a factor too early: in $\frac{\sin x - x}{x^3}$ you need $\sin x$ to x^3 , not x .
- ▶ Composing when $g(0) \neq 0$: $e^{\cos x} = e \cdot e^{\cos x - 1}$; expand in $\cos x - 1 = O(x^2)$.
- ▶ Forgetting cross terms in $(a + b + c)^2$.
- ▶ Using a series outside its radius ($\ln(1 + x)$ at $x = 2$ diverges).

7 Multivariable Taylor: the formulas

Write $\mathbf{a}$ for the base point and $\mathbf{h} = \mathbf{x} - \mathbf{a}$. In two variables, $\mathbf{h} = (h, k)$.

Operator form (any dimension)

$$f(\mathbf{a} + \mathbf{h}) = \sum_{m=0}^n \frac{1}{m!} (\mathbf{h} \cdot \nabla)^m f(\mathbf{a}) + R_n$$

$$R_n = \frac{1}{(n+1)!} (\mathbf{h} \cdot \nabla)^{n+1} f(\mathbf{a} + \theta \mathbf{h}), \quad 0 < \theta < 1$$

Proof idea: apply the 1-D theorem to $g(t) = f(\mathbf{a} + t\mathbf{h})$ at $t = 1$.

Then $g^{(m)}(0) = (\mathbf{h} \cdot \nabla)^m f(\mathbf{a})$.

Second order in matrix form

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a})^\top \mathbf{h} + \frac{1}{2} \mathbf{h}^\top H(\mathbf{a}) \mathbf{h}$$

with Hessian $H_{ij} = \partial_i \partial_j f$ (symmetric when $f \in C^2$).

Two variables, written out (all derivatives at (a, b)):

$$\begin{aligned} f(a+h, b+k) &= f + (hf_x + kf_y) \\ &+ \frac{1}{2!} (h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) \\ &+ \frac{1}{3!} (h^3 f_{xxx} + 3h^2 k f_{xxy} \\ &+ 3hk^2 f_{xyy} + k^3 f_{yyy}) + \dots \end{aligned}$$

The coefficients 1, 2, 1 and 1, 3, 3, 1 are **binomial** because $(h\partial_x + k\partial_y)^m$ expands like $(h+k)^m$ (mixed partials commute).

Multi-index form

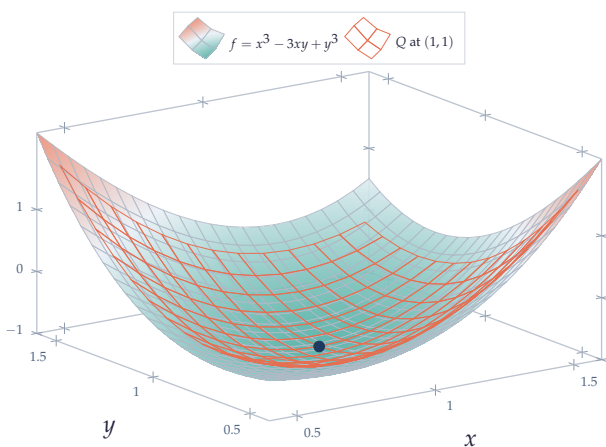
With $\alpha = (\alpha_1, \dots, \alpha_d)$, $|\alpha| = \sum \alpha_i$, $\alpha! = \prod \alpha_i!$, $\mathbf{h}^\alpha = \prod h_i^{\alpha_i}$:

$$f(\mathbf{a} + \mathbf{h}) = \sum_{|\alpha| \leq n} \frac{D^\alpha f(\mathbf{a})}{\alpha!} \mathbf{h}^\alpha + R_n$$

Read off derivatives: the coefficient of $h^i k^j$ is $\frac{\partial_x^i \partial_y^j f}{i! j!}$.

Counting terms. Degree- m terms in d variables: $\binom{m+d-1}{d-1}$. In 2-D: $m+1$. In 3-D: $\frac{(m+1)(m+2)}{2}$.

8 Picture: surface vs. quadratic model



The quadratic model Q (coral mesh) sits tangent to f at its minimum $(1, 1, -1)$ and shares its curvature.

9 Worked multivariable examples

Example 2.1 By products: $e^x \sin y$ at $(0, 0)$ to order 3

Multiply the 1-D series, keeping total degree ≤ 3 :

$$\left(1 + x + \frac{x^2}{2}\right) \left(y - \frac{y^3}{6}\right)$$

Keep $y, xy, \frac{x^2 y}{2}, -\frac{y^3}{6}$ (drop xy^3 , etc.)

Result $e^x \sin y \approx y + xy + \frac{1}{2}x^2 y - \frac{1}{6}y^3$

Bonus: the $x^2 y$ coefficient is $\frac{f_{xxy}}{2!1!}$, so $f_{xxy}(0, 0) = 1$ with no differentiation.

Example 2.2 By derivatives: $f = x/y$ at $(1, 1)$ to order 2

Derivatives $f = 1, f_x = \frac{1}{y} = 1, f_y = -\frac{x}{y^2} = -1$
 $f_{xx} = 0, f_{xy} = -\frac{1}{y^2} = -1, f_{yy} = \frac{2x}{y^3} = 2$

Assemble with $h = x - 1, k = y - 1$:

$$1 + h - k + \frac{1}{2}(0 \cdot h^2 + 2(-1)hk + 2k^2)$$

$$\frac{x}{y} \approx 1 + h - k - hk + k^2$$

Cross-check by series $\frac{1+h}{1+k} = (1+h)(1-k+k^2-\dots) = 1+h-k-hk+k^2+\dots \checkmark$

Use it $\frac{1.1}{0.95}: h = 0.1, k = -0.05 \Rightarrow 1.1575$ (true 1.15789...).

Example 2.3 Substitution: $\sqrt{1+x+2y}$ at $(0, 0)$ to order 2

Let $u = x + 2y$ and use $\sqrt{1+u} = 1 + \frac{u}{2} - \frac{u^2}{8}$:

$$1 + \frac{x+2y}{2} - \frac{x^2+4xy+4y^2}{8}$$

$$\approx 1 + \frac{x}{2} + y - \frac{x^2}{8} - \frac{xy}{2} - \frac{y^2}{2}$$

Linear combinations inside a 1-D function: expand in u , then multinomial-expand u^m .

Example 2.4 Substitution in the centre: $e^x \ln y$ at $(0, 1)$

Put $y = 1 + k$: $\ln(1+k) = k - \frac{k^2}{2} + \dots$ and $e^x = 1 + x + \dots$. To order 2:

$$(1+x) \left(k - \frac{k^2}{2}\right) \Rightarrow k + xk - \frac{k^2}{2}, \quad k = y - 1$$

So $f = 0, \nabla f = (0, 1), f_{xy} = 1, f_{yy} = -1, f_{xx} = 0$ at $(0, 1)$.

Example 2.5 Three variables: $e^{x+2y} \cos z$ at 0

With $u = x + 2y$: $(1 + u + \frac{u^2}{2})(1 - \frac{z^2}{2}) \approx 1 + u + \frac{u^2}{2} - \frac{z^2}{2}$.

$$= 1 + x + 2y + \frac{1}{2}(x^2 + 4xy + 4y^2) - \frac{1}{2}z^2$$

Read off gradient and Hessian:

$$\nabla f(0) = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad H(0) = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Check: $\frac{1}{2} \mathbf{h}^\top H \mathbf{h} = \frac{1}{2}(x^2 + 4xy + 4y^2 - z^2)$. The off-diagonal 2 appears *twice* in the quadratic form, giving $4xy/2$.

10 Critical points: the Hessian test

At a critical point $\nabla f(\mathbf{a}) = \mathbf{0}$, so the linear term vanishes and

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \approx \frac{1}{2} \mathbf{h}^\top H \mathbf{h}.$$

The **sign pattern of the quadratic form** decides the shape:

Classification

Eigenvalues of H all > 0 local **minimum**
 Eigenvalues all < 0 local **maximum**
 Mixed signs **saddle**
 Some $= 0$, rest one sign **inconclusive**: go to order 3+

2-D shortcut: $D = f_{xx}f_{yy} - f_{xy}^2$.
 $D > 0, f_{xx} > 0$: min. $D > 0, f_{xx} < 0$: max. $D < 0$: saddle.

Example 3.1 $f = x^3 - 3xy + y^3$

Critical points $f_x = 3x^2 - 3y = 0, f_y = 3y^2 - 3x = 0 \Rightarrow y = x^2, x = x^4$
 so $(0,0)$ and $(1,1)$.

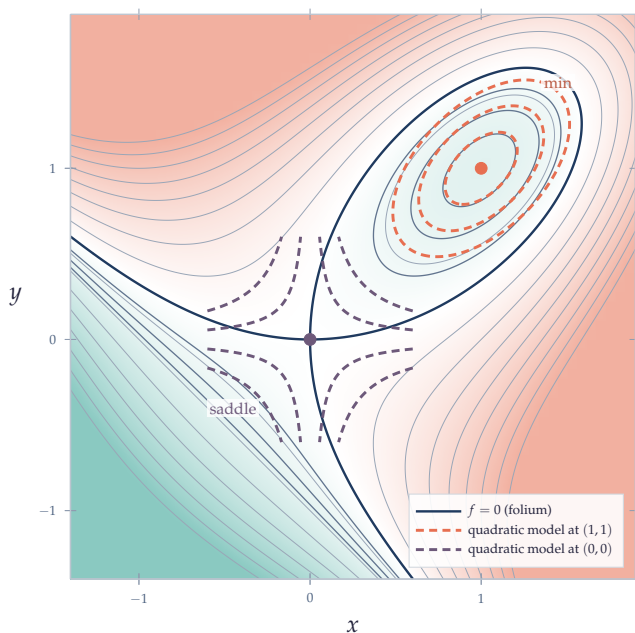
Hessian $H = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$.

At $(1,1)$ $D = 36 - 9 = 27 > 0, f_{xx} = 6 > 0$: **minimum**, and

$$f \approx -1 + 3(h^2 - hk + k^2)$$

(positive-definite form: level sets are **ellipses**).

At $(0,0)$ $D = -9 < 0$: **saddle**, $f \approx -3xy$ (level sets are **hyperbolas**).

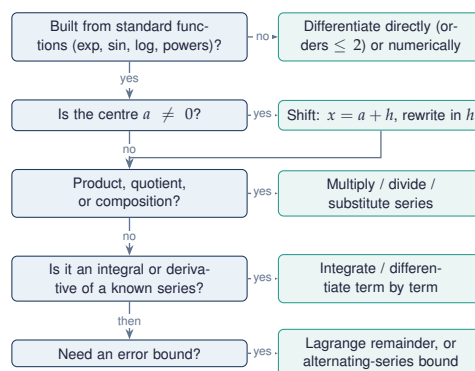


Level curves of f (solid) vs. its Taylor quadratics (dashed). Near each critical point the true contours converge onto the model's ellipses or hyperbolas.

When the quadratic test fails

$f = x^2 + y^4$ and $g = x^2 - y^4$ both have $H = \text{diag}(2, 0)$ at $\mathbf{0}$, yet f has a minimum and g a saddle. A zero eigenvalue means the **quartic terms** decide: look along the degenerate direction.

11 Which method? A decision map



12 Little-o / big-O arithmetic

As $x \rightarrow 0$, for $m \leq n$:

- ▶ $O(x^m) + O(x^n) = O(x^m)$ (the lower power wins)
- ▶ $x^m \cdot O(x^n) = O(x^{m+n})$, $O(x^m) O(x^n) = O(x^{m+n})$
- ▶ $\frac{O(x^n)}{x^m} = O(x^{n-m})$ (dividing *loses* precision: expand further first)
- ▶ $c O(x^n) = O(x^n)$; the same rules hold for $o(\cdot)$

Multivariable: use $O(\|\mathbf{h}\|^n)$, where every monomial $h^i k^j$ with $i + j = n$ is $O(\|\mathbf{h}\|^n)$.

13 Recipe: a multivariable expansion in 5 steps

1. Fix the order n and centre $\mathbf{a}$; set $\mathbf{h} = \mathbf{x} - \mathbf{a}$.
2. If f is a product or composition of 1-D functions, expand each and multiply, **dropping every monomial of total degree $> n$** .
3. Otherwise compute all partials up to order n at $\mathbf{a}$ (use symmetry: $f_{xy} = f_{yx}$).
4. Assemble with binomial weights: $\frac{1}{m!} \sum \binom{m}{i} \partial_x^i \partial_y^{m-i} f h^i k^{m-i}$.
5. Sanity-check one number: plug a small $\mathbf{h}$ into both f and T_n .

14 Quick self-test

- Q1.** $\ln(\cos x)$ to $O(x^6)$.
Q2. $\lim_{x \rightarrow 0} \frac{1 - \cos x - \frac{x^2}{2} \cos x}{x^4}$.
Q3. $\cos(x + y^2)$ at $\mathbf{0}$ to order 4.
Q4. Classify $\mathbf{0}$ for $f = x^2 + 3xy + y^2$.

Answers

- Q1** $-\frac{x^2}{2} - \frac{x^4}{12}$ (compose $\ln(1 + u)$, $u = \cos x - 1$).
Q2 numerator $= (\frac{x^2}{2} - \frac{x^4}{24}) - \frac{x^2}{2} (1 - \frac{x^2}{2}) = \frac{x^4}{4} - \frac{x^4}{24}$, so $\frac{5}{24}$.
Q3 $1 - \frac{x^2}{2} - xy^2 + \frac{x^4}{24} - \frac{y^4}{2}$.
Q4 $D = 4 - 9 < 0$: saddle.